

Auditable Long-Term Memory: A Deterministic Retrieval Chain at $95.4 \pm 0.4\%$ on LongMemEval-S

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Abstract

We report 479/500 and 475/500 across two independent full passes on LongMemEval-S (Claude Opus reader; CLI alias resolving to `claude-opus-5`, §5.3) under the benchmark’s official GPT-4o judge, against a published state of the art of 478/500 (95.60%, Chronos High, PwC, arXiv 2603.16862) — one pass above it, one below: a statistical tie whose spread sits inside the instrument’s measured noise band. A second, grok-4.6 reader configuration measured 476/474 on the identical substrate. The system is a deterministic retrieval chain — dense retrieval, cross-encoder reranking, a coverage-first packet compiler, and mechanically extracted reasoning scaffolds, all deterministic code validated by negative controls — with a replaceable LLM reader confined to the final answering step. At margins of a few questions, measurement is the result: the pair’s entire spread traced to eight verdict-flip rows; cross-judge agreement is 98%; we observed official-judge verdict flips on byte-identical answers; and a maximum-reasoning-effort reader variant scored *lower* (461/465), with 4 of its 19 per-pass losses being the same answer text judged oppositely. We release every reader output, judge verdict, and control receipt, plus an error dossier that attributes all remaining failures — including two strict gold-answer defects, an improvement stage rejected by our own negative control, and the routes that did not work. A full re-score under the official judge costs about \$1.28.

1 Introduction

Long-context memory benchmarks are typically reported as single numbers from single runs, judged by an LLM, with no disclosure of run-to-run or judge variance. At the top of a leaderboard, where entries are separated by a handful of questions out of 500, this practice makes the ranking partially an artifact of measurement noise.

This report makes two claims. **A capability claim:** a deterministic, auditable retrieval chain — in which the language model is confined to a replaceable reader role — measures 479/475 per 500 on LongMemEval-S, pass 1 above the published state of the art, the pair a statistical tie with it, with overlapping confidence intervals and per-run variance comparable to the gap. **A methodology claim:** the apparatus that produced the number is the more important contribution. Every stage below the reader is deterministic code validated by negative controls; the promotion rule was frozen before the deciding run; measurement noise from both the reader and the judge is quantified and disclosed rather than absorbed into the headline.

The system under test is the retrieval core of Archivist, a local-first conversational memory product.

Nothing in the chain is benchmark-specific except the evaluation harness itself.

1.1 Related work

Agentic memory systems (MemGPT, Packer et al., 2023; production systems such as Mem0 and Zep) focus on memory *stores* and management policies; our contribution is orthogonal — determinism below the reader, with the store fixed. Long-term memory benchmarks include LongMemEval (Wu et al., 2024), used here, and LoCoMo (Maharana et al., 2024); we evaluate one benchmark and claim no cross-benchmark generality (§7). The retrieval stack follows the retrieve-and-rerank tradition (Nogueira & Cho, 2019; BGE-M3 family, Chen et al., 2024); reciprocal rank fusion (Cormack et al., 2009) is among the extension-ordering baselines swept in §3.3. On measurement: a self-consistency step (Wang et al., 2023) was built and later withdrawn when internal review showed its two-pass framing was a derived identity (§4.4); the judge-variance study operationalizes known LLM-as-judge concerns (Zheng et al., 2023); the two-pass reporting rule answers the single-run critique of Dodge et al. (2019).

2 Benchmark and evaluation protocol

LongMemEval-S (Wu et al., 2024) contains 500 questions over long multi-session chat histories: 470 answerable questions across six types (single-session-user, single-session-assistant, single-session-preference, multi-session, temporal-reasoning, knowledge-update) and 30 abstention questions whose correct behavior is to decline to answer. Evidence for a question may be spread across sessions recorded weeks apart.

Judge. We use the benchmark’s official protocol: GPT-4o (gpt-4o-2024-08-06) with the paper’s rubric templates. One header sentence of the single-session-preference template was reconstructed from the benchmark’s published rubric text; the templates as used are SHA-256 pinned (9c9d67fab129...) and released verbatim, so the reconstruction is auditable rather than asserted. Every scoring run executes 12 live positive and 12 live negative control verdicts against a 0.9 pass threshold (preference rows are excluded from the positive control arm because their rubric is the reconstructed one); every run in this report returned 12/12 and 12/12. The harness records the resolved model snapshot per verdict and fails closed on authentication errors. A full 500-row re-score costs $\approx$ \$1.28.

Comparison point (verified at source, 2026-08-31). The published raw-score state of the art is **Chronos High: 478/500 (95.60%)** (PwC, arXiv 2603.16862, Mar 2026; generator Claude Opus 4.6, LongMemEval judge protocol, per-category table published — their Table 2 and Appendix A, Table 4). Two widely-quoted higher percentages are not raw-comparable: Mastra OM’s “94.87%” is a category-unweighted average — their own per-category counts sum to **468/500 raw** (generator gpt-5-mini) — and OMEGA’s “95.4%” is likewise task-averaged with raw 466/500, judged by GPT-4.1 rather than the canonical judge. One higher claim (481/500, “agentmemory V4”) exists in an unreviewed personal repository; we note it without treating it as the published bar. Reader tier varies freely across published entries (GPT-4o through Opus 4.6; our headline reader resolves to Opus 5, §5.3); we follow the field convention of reporting our strongest configuration alongside the full reader ladder.

3 System architecture

The chain has five stages. Stages 1–4 are deterministic: byte-identical outputs across runs are enforced by verification gates, not assumed (verified on the pinned environment; hardware and library versions are recorded in the release).

3.1 Dense retrieval. Per-question dense embedding retrieval over the haystack produces a ranked candidate pool (local inference; pool ceiling: 468/470 questions have all gold sessions somewhere in the pool).

3.2 Cross-encoder reranking. A cross-encoder (BAAI/bge-reranker-v2-m3) scores each candidate session against the question; session score is the maximum chunk score (chunk = date header + turn). Scoring is rank-deterministic (fp16, fixed batch, tie-break by prior rank).

3.3 Coverage-first packet compilation. For each question a reading packet is compiled under a hard budget (16 sessions / 40k proxy tokens): the dense baseline top-10 is preserved as a byte-identical prefix, remaining slots fill in cross-encoder order with term-preserving extractive compression, and budget overruns drop the lowest-ranked extension first. Validated by determinism, gold-permutation and question-ID-sabotage negative controls, and budget accounting. Packets are gold-complete for 462/470 questions; a sweep of all five extension-ordering policies confirmed the shipped ordering is globally optimal (462 vs. 454 best alternative).

3.4 Deterministic reasoning scaffolds (“facts”). For 240 of 470 questions the chain emits a deterministic evidence index computed from the packet: session chronology, a canonical user-statement recency domain with clock-granular tie keys, a value-preference invariant, candidate enumeration with quantity roll-up for counting questions, and granularity/unit and event-anchor rules (contract v3.4-facts-20260831). Scaffolds are code with in-code coupling invariants that raise on internal inconsistency. The scaffold is gold-blind and question-ID-independent by construction.

3.5 Replaceable reader. The packet, scaffold, and question go to an LLM reader, deliberately replaceable; we measure the chain under multiple readers (§5.3).

4 Measurement methodology

4.1 Two-pass rule (pre-registered). A nondeterministic reader means a single full run at a small margin is not a reportable number. Our promotion rule, adopted before the deciding experiments: a headline requires two full passes agreeing, or one number with a stated $\pm$ spread from measured flip variance.

4.2 Judge variance is measured, not assumed. We re-judged both passes of our prior configuration with a second judge on identical rows: agreement 98.2%/97.8%, the second judge strictly harsher. We also observed the official judge returning opposite verdicts on a byte-identical answer across runs. At these margins the judge is the measurement floor; we log such rows rather than resolve them by re-rolling.

4.3 Negative controls at the improvement layer. Every proposed improvement runs against a 60-row control set of baseline-correct questions before integration. This rule has teeth: a verifier stage that recovered 3 previously-wrong rows was *rejected* because the control replay showed it flipped 11 of 59 correct drafts to wrong (§6.2). The reader-tier upgrade in this report passed the same control with zero flips before its full pair was launched.

4.4 Self-consistency tie-break (built, then withdrawn). The raw pair’s spread came from eight verdict-flip rows, and a tie-break rule frozen before any additional read gave each flip row three more reads with a five-vote majority. The resulting number was WITHDRAWN from any claim: internal review showed the construction wrote one derived verdict into both passes (an identity, not two agreeing measurements) and its flip-row selection was judge-conditioned. The artifacts remain released as a record; no self-consistency number is claimed anywhere in this report.

4.5 What we did not do. No per-question prompt tuning against gold; no judge re-rolling; no selective reporting (failed routes are in §6); no counting of gold-defect adjudication in any headline number.

5 Results

5.1 Headline

Configuration	Pass 1	Pass 2
v3.4, Claude Opus reader (CLI lane)	479/500	475/500
v3.4, grok-4.6 high (raw pair)	476/500	474/500
v3.4, grok-4.6 xhigh (agentic transport)	461/500	465/500

The Opus pair is the headline: pass 1 exceeds the published SOTA (478) on the raw score; the pair’s both-pass-stable floor is 473 (vs. the grok pair’s 471); pass 1 is perfect on three of six answerable types (single-session assistant, user, preference). Its 8 flip rows (6 lost — including 2 abstentions and 2 dossier-listed flip-prone rows — 2 gained) match the flip-noise structure of every other pair here. Under the pre-registered two-pass rule this is a **statistical tie with the published SOTA, one pass above it**, not a clean beat: we state 477 ± 2 and decline the stronger claim our own bar forbids (a pre-release re-judge of pass 1 under the same official judge returned 478 — three verdict flips on identical text, §8 — so even “one above” is a judge roll). The two gold-defect rows from the dossier (§6.3) sit inside this pair as well: one is scored correct via the fabrication the dossier documents (the reader supplies the \$500 no session contains), one is lost to fidelity (the evidence-faithful answer contradicts defective gold) — net zero per pass. Under adjudication the pair reads 478/474 (ex-fabrication) to 480/476 (restored); the tie with the published state of the art holds in every stance. Judge controls 24/24 in both passes; a 25-row probe preceded the pair (11/25 stable-wrong recovered, one row no prior reader had ever solved).

The xhigh variant is a measured regression ($-15/-9$), reported in full. Its gate evidence had looked favorable — a 25-row probe on the high pair’s wrong set recovered 9, and a 60-row negative control on baseline-correct rows showed zero flips — but the full pair exposed the gates’ limits: only 4 probe recoveries held, and the true correct-row flip rate ($\sim 3.4\%$) was small enough that a 60-row control had a $\sim 13\%$ chance of showing zero flips. Row-level attribution of pass 1’s 19 losses: 4 are judge flips on answer text identical to the high run’s, and 15 share one over-skepticism signature — wrong abstentions on answerable rows, over-filtered counts, hedged golds. More reasoning effort made the reader second-guess correct evidence. (One v1 loss was a staleness artifact, not a reader loss: the reader resumed one row eleven minutes after the judge had scored it on an empty response; a surgical v2 re-judge under fresh 12/12 controls restored it, $460 \rightarrow 461$.)

Prior-configuration detail: answerable 450/470 and 447/470; abstention 26/30 and 27/30. Judge controls passed in every scoring run. Under the gold-defect adjudication scenario (§6.3) the prior pair reads 478/476; we report that as a scenario, never as a score.

5.2 Per-question-type breakdown (official GPT-4o judge)

Question type	<i>n</i>	Raw p1	Raw p2	xhigh p1 / p2	Opus p1 / p2
single-session-assistant	56	56 (100.0%)	56 (100.0%)	55 / 55	56 / 56
single-session-user	64	63 (98.4%)	63 (98.4%)	61 / 60	64 / 64
knowledge-update	72	70 (97.2%)	71 (98.6%)	69 / 70	69 / 70
single-session-preference	30	29 (96.7%)	29 (96.7%)	25 / 25	30 / 28
temporal-reasoning	127	122 (96.1%)	122 (96.1%)	121 / 123	123 / 123
multi-session	121	110 (90.9%)	106 (87.6%)	103 / 106	110 / 109
abstention	30	26 (86.7%)	27 (90.0%)	26 / 26	27 / 25
Total	500	476	474	461 / 465	479 / 475

Five of six answerable types are pass-stable to within one question; the raw pair’s entire answerable spread is one type (multi-session), partially offset by knowledge-update. Against the published SOTA’s category table on its own abstention-folded basis (*n*: KU 78, MS 133, SSA 56, SSP 30, SSU 70, TR 133 — their Appendix A), our pass-1 gap is -3 knowledge-update and -1 preference, partially offset by $+2$ multi-session (120/133 vs. their 118/133); pass 2 trails on multi-session as well (116/133), so our multi-session score exceeds theirs in pass 1 only.

5.3 Ablation ladder (single ruler: official GPT-4o judge)

Stage	Score /500
Dense retrieval + flat reading (Flash reader, v2 prompts)	452/499 ¹
+ chain, v3.3 operators, Sol reader	468
+ cross-encoder rerank (v3, grok reader)	473
+ v3.4 operators (official pair, grok high)	476 / 474
+ reader xhigh + agentic transport	461 / 465 (regression)
+ reader tier: Claude Opus, same substrate	479 / 475

Ex-dossier (all eight gold-defect/boundary rows removed from both readers): grok leads 475/474 vs. Opus 474/471 — Opus’s net margin over grok is carried by gold-idiom reading on dossier rows, not by evidence reading on ordinary rows. The Opus reader is the CLI alias `opus` (claude-cli lane, subscription), not a pinned snapshot: no run artifact records the run-time resolution, and a same-day probe (2026-09-01, Claude Code CLI 2.1.258; the runs used 2.1.252) resolved `opus` to `claude-opus-5` — one model generation newer than the Chronos comparator’s Opus 4.6. The grok and GLM readers are named models. Three of Opus’s abstention credits (2133c1b5_abs, c8090214_abs, f685340e_abs) are premise-repair-then-answer responses the official judge accepts and the GLM cross-judge rejects; under strict abstention the pair reads 476/472 — still a tie. ¹One row of the legacy baseline failed its reader lane and was never re-read; counted as scored-out-of-499 rather than imputed.

5.4 Reader robustness

On identical packets and scaffolds: **Claude Opus 479/475**; grok-4.6-high 476/474; GPT-5.6 (Sol) 455 — measured decomposition versus its prior-contract 468: -5 per-run reader variance on byte-identical non-scaffold rows, -5 negative scaffold $\times$ reader interaction, -3 abstention; judge controls 24/24. DeepSeek V4 Pro scores 436/500 (407/470 answerable) with the best abstention behavior of any reader measured (29/30). A 30B floor probe (Nemotron-3.5-Lightning) collapses to 93/500 on the identical gold-complete packets — the chain does not rescue a reader below a capability threshold. We controlled for a serving artifact: correctness barely varies with packet size (28% vs. 15% across size buckets, vs. 90/86% for V4 Pro), and a question-first re-prompt of a failed row

was still ignored in favor of distractor-session content — the collapse is model-side, not provider truncation. A gemini-3.7-flash run reached a 189-row partial (a type-ordered prefix, not a sample: single-session-user 64/64, multi-session 73/83, preference 21/30; no other types) and is reported as a bounded signal only. A GLM-5.3 run on the identical substrate scores 471/500 (445/470 answerable, 26/30 abstention; official GPT-4o judge, controls 12/12) — a single-pass curve datapoint per the two-pass rule, not a headline. A Kimi K3 run (the long-context-specialist family) scores 465/500 (444/470 answerable, 21/30 abstention; same judge, controls 12/12) — long-context specialization does not transfer to memory-benchmark dominance on this substrate. Together the fixed substrate exposes a full reader-capability curve (93 → 436 → 455 → 465 → 471 → 476): the packet and scaffold layers carry the score for capable readers, and reader choice moves it by tens of points across tiers but only a few points within a tier. The v3.4 scaffold contract is tuned to the grok reader family (§7).

6 Error analysis: every remaining failure, attributed

Of 470 answerable questions, 24 fail in at least one prior-configuration pass.

6.1 Reader-cognition residue (~4 rows). The packet contains the correct evidence; the reader commits to an evidence-backed distractor. The residue is question-boundary semantics (does a stated *plan* update a location? does the offered property count as “viewed?”), not evidence selection.

6.2 The verifier our own control rejected. A cite-or-revise verifier recovered 3/12 winnable wrong rows — and the mandatory negative control showed it revised 19/59 baseline-correct drafts, flipping 11 to wrong while repairing none. Blocked by the pre-committed acceptance rule. A second design (draft-blind candidate enumeration + rule-based adjudication) recovered 0/9. Both failures are published; they map the limit of prompting-side repair.

6.3 Gold-answer defects (2 strict, 6 boundary). Documented row-by-row in the released dossier. Strict A: gold requires a \$500 expense that appears nowhere in the evidence, plus an event dated after the question date attended in the past tense; a weaker reader that *hallucinated* the missing \$500 was scored correct by the official judge. The grok reader’s evidence-faithful \$220 scores 0 in both passes; we keep the 0. The headline Opus reader, by contrast, supplies the \$500 and is scored correct on both passes — the disclosed fabrication point inside 479/475 (§5.1). Strict B: gold says “15 weeks” between two user-dated events 81 days apart. Six further rows are boundary-semantic and claimed as worth nothing. The Chronos authors independently document a defect on question 6d550036 (their §3.5, alongside a judge-variability example on 75f70248), consistent with our dossier.

6.4 Retrieval ceiling and coverage, measured precisely. One row’s gold never reaches the candidate pool; three rows lost an in-pool gold session to packet budget ordering (no global ordering recovers them without losing more). An entity-linking retrieval arm was built, tested (17/17), and retired when measurement showed zero of its added sessions were gold. Remaining failures are measured flip noise.

7 Limitations

- **Margins at this level are not statistical superiority.** Wilson 95% intervals: our 479 [93.7%, 97.2%] and 475 [92.7%, 96.6%] (grok pair: 476 [93.0%, 96.8%], 474 [92.5%, 96.4%]) vs. Chronos’s 478 [93.4%, 97.1%] — overlapping almost entirely. (The xhigh pair sits lower: 461 [89.5%, 94.2%], 465 [90.4%, 94.9%].) Nearby entries’ intervals overlap substantially;

no comparison entry publishes per-question verdicts, so McNemar is not computable. Our number is reproduced with variance disclosed; comparison entries are single runs.

- **The judge is the floor.** $\sim 2\%$ verdict variance between judges, and observed flips on identical answers, bound what any entry at these margins can honestly claim.
- **Scaffold–reader coupling.** The v3.4 scaffold is tuned to one reader family; the exact number moves a few points across readers, though the chain’s ranking of readers is stable.
- **One benchmark.** No cross-benchmark generality claim.

8 Reproducibility

Released with this report at <https://github.com/cjchanh/longmemeval-evidence> (MIT). A fresh clone verifies 448/448 manifest hashes and re-derives every headline count from the released verdicts without an API key. The evidence commits in the footer anchor to the source tree the release was exported from. The release holds all reader outputs, all judge verdicts and control receipts for every run in the ladder, the judge harness (frozen rubric, SHA-pinned templates, fail-closed transport), the tie-break rule and per-vote records, attribution tables, the gold-defect dossier, and run scripts. Re-scoring any run under the official judge costs $\approx \$1.28$ and requires only an OpenAI API key. The released/held boundary is frozen in `RELEASE_MANIFEST.md` (one SHA-256 per released file, regenerated deterministically by `scripts/build_release_manifest.py`): released are all reader checkpoints, judge verdicts, control receipts, agreement matrices, the dossier, the judge harness with its tests and rubric source, and the reader-lane scripts; held are the retrieval, rerank, packet-compiler and scaffold-operator sources and the materialized packets (benchmark haystack text). The held stages ship as pinned artifacts with verification receipts (determinism, gold-permutation, question-ID-sabotage, budget gates); every judge-side claim is re-derivable from the released checkpoints and harness alone. The benchmark data is MIT-licensed (`xiaowu0162/longmemeval-cleaned`), so the released checkpoints may carry question and gold text. Judge-side reproduction, run before release: the frozen pass-1 reader checkpoint was re-judged the same day in a clean output directory with the identical harness (`gpt-4o-2024-08-06`, resolved snapshot exact, fresh controls 12/12 and 12/12). Result **478/500**: 497 of 500 verdicts agree with the quoted 479; answerable is identical (452/470); the three flips are on byte-identical answer text (7024f17c and 031748ae_abs lost, a2f3aa27 gained). The official judge’s own flip floor ($\$4.2$) is therefore measured within a single judge on the headline pass, and “one pass above” is inside that floor; the tie is the claim. Receipts: `full-v34-opus_pass1/judge_gpt4o_repro_20260901/`.

9 Conclusion

A deterministic, negatively-controlled retrieval chain with the LLM confined to a replaceable reader measures 479/475 per 500 on LongMemEval-S — one pass above the published state of the art, the pair a statistical tie inside the instrument’s noise band — under a measurement discipline that makes the number defensible at margins where measurement is usually the weakest link. The apparatus generalizes: freeze the promotion rule before the deciding run, control every improvement against what it might break, measure the judge, and publish the failures with the successes.

References

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